

SCAPPOOSE, OR, USA

WMO: 726836

Lat: 45.773N

Lon: 122.861W

Elev: 17

StdP: 101.12

Time Zone: -8.00 (NAP)

Period: 01-19

WBAN: 04201

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-5.2	-3.3	-10.2	1.6	-1.9	-7.3	2.0	-0.8	7.4	8.6	6.0	8.8	1.1	260	0.493

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	14.4	33.4	19.8	31.1	19.2	28.9	18.4	20.8	30.7	19.9	29.3	18.9	27.6	3.7	340

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.3	12.4	24.2	16.3	11.6	22.8	15.5	11.0	22.1	60.1	30.8	56.6	29.2	53.5	27.8	25.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
6.4	5.5	4.9	DB	-8.4	37.7	2.6	1.8	-10.3	39.0	-11.8	40.1	-13.2	41.1	-15.1	42.4	(4)
			WB	-9.1	23.1	2.4	1.1	-10.8	23.9	-12.3	24.6	-13.6	25.2	-15.4	26.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	11.6	4.4	5.6	7.9	10.2	13.9	16.7	20.0	20.2	17.0	11.8	7.1	4.1	(6)
		DBStd	6.49	3.42	2.86	2.77	2.99	3.38	3.40	3.01	2.96	3.23	2.89	3.34	3.46	(7)
		HDD10.0	711	174	123	78	32	6	0	0	1	15	97	186		(8)
		HDD18.3	2643	431	355	324	244	143	73	15	13	63	204	336	443	(9)
		CDD10.0	1299	1	2	12	38	127	199	311	317	210	70	11	1	(10)
		CDD18.3	191	0	0	0	1	6	22	68	71	23	1	0	0	(11)
		CDH23.3	2578	0	0	1	29	137	325	850	868	351	17	0	0	(12)
		CDH26.7	1018	0	0	0	5	39	124	356	369	124	2	0	0	(13)
(14)	Wind	WSAvg	1.8	1.3	1.4	1.8	1.9	2.0	2.2	2.7	2.4	1.9	1.4	1.4	1.4	(14)
(15)	Precipitation	PrecAvg	1312	208	129	132	106	75	50	13	22	39	111	208	225	(15)
(16)		PrecMax	1947	420	333	230	223	151	104	58	88	99	207	455	449	(16)
(17)		PrecMin	1011	73	0	35	30	0	12	0	0	26	71	119		(17)
(18)		PrecStd	242	78	73	48	54	41	22	13	20	27	49	96	85	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	DB	13.9	16.3	21.5	27.0	30.3	34.1	36.9	36.4	33.4	25.9	17.5	14.0		(19)
(20)		MCWB	11.6	9.9	12.8	16.0	17.2	21.1	21.0	19.9	18.9	15.4	13.8	12.2		(20)
(21)		DB	12.3	13.9	17.8	22.7	27.3	30.2	33.0	33.4	30.1	22.0	15.3	11.8		(21)
(22)		MCWB	10.2	8.9	11.2	13.8	16.8	18.9	19.7	19.8	17.7	14.7	12.8	10.2		(22)
(23)		DB	10.8	12.2	15.2	18.9	23.9	27.1	30.6	30.9	27.5	19.3	13.6	10.2		(23)
(24)		MCWB	9.2	8.7	10.1	11.9	15.5	17.7	19.4	19.5	17.4	13.9	11.2	8.7		(24)
(25)		DB	9.3	10.6	13.0	16.3	21.1	24.0	28.0	28.0	24.7	17.1	12.4	8.7		(25)
(26)		MCWB	7.8	8.1	9.5	10.9	14.2	16.3	18.5	18.5	16.5	13.3	10.5	7.4		(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	WB	12.6	11.7	14.1	16.8	19.2	21.6	22.8	21.9	20.7	17.8	15.1	12.6		(27)
(28)		MCDB	13.6	13.4	19.8	25.9	28.3	31.9	34.3	32.8	29.6	22.0	16.4	13.8		(28)
(29)		WB	10.7	10.6	12.2	14.3	17.5	19.6	20.8	20.7	19.0	15.9	13.1	10.6		(29)
(30)		MCDB	11.5	12.6	15.5	21.0	25.6	29.0	30.3	30.7	27.3	19.6	14.7	11.5		(30)
(31)		WB	9.2	9.3	11.1	12.8	16.1	18.0	19.8	19.8	17.9	14.8	11.9	8.9		(31)
(32)		MCDB	10.3	11.2	14.1	17.6	22.7	26.0	29.1	29.4	25.4	17.9	13.0	9.9		(32)
(33)		WB	8.0	8.2	10.1	11.5	14.8	16.8	18.8	18.8	16.9	13.8	10.7	7.7		(33)
(34)		MCDB	9.0	10.1	12.9	15.3	20.4	23.3	27.5	27.4	23.3	16.5	12.0	8.5		(34)
(35)	Mean Daily Temperature Range	MDBR	6.6	8.5	9.6	10.7	11.4	11.7	13.9	14.4	13.4	10.4	7.9	6.0		(35)
(36)		5% DB	8.1	11.8	13.8	16.7	17.7	17.3	18.3	18.8	18.5	14.9	9.4	7.1		(36)
(37)		MCWBR	6.1	7.5	8.1	8.9	9.0	8.2	7.8	7.7	8.2	8.3	6.4	5.5		(37)
(38)		5% WB	6.8	8.7	11.1	14.2	15.3	15.9	16.7	17.3	16.0	11.4	7.1	6.1		(38)
(39)		MCWBR	5.5	6.0	7.0	8.1	8.2	7.9	7.7	7.7	7.6	7.0	5.4	5.2		(39)
(40)	Clear-Sky Solar Irradiance	taub	0.321	0.316	0.333	0.361	0.377	0.365	0.350	0.357	0.343	0.332	0.325	0.327		(40)
(41)		taud	2.525	2.551	2.514	2.398	2.372	2.421	2.488	2.479	2.507	2.548	2.525	2.463		(41)
(42)		Ebn at Noon	789	872	902	898	890	901	910	892	877	839	776	731		(42)
(43)		Edh at Noon	66	76	91	111	118	113	104	101	90	76	65	63		(43)
(44)	All-Sky Solar Radiation	RadAvg	1.14	1.98	2.88	4.20	5.34	5.90	6.69	5.72	4.27	2.47	1.33	0.93		(44)
(45)		RadStd	0.19	0.28	0.38	0.39	0.51	0.56	0.48	0.39	0.38	0.31	0.18	0.14		(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+113	N/A
(47)	Regional (35 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+96	+28

Nomenclature: See separate page